단순선형회귀

선형성: β에 대한 선형

정규성: $Y|X=x$은 정규분포

등분산성: $\mathbb{E}(Y|X=x)$의 분산은 상수

독립성: 측정오차 ϵ은 서로 독립

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(o, \sigma^2)$

$\hat\beta_0 + \hat\beta_1 \bar x = \bar y, \quad \hat\beta_0 \sum x_i + \hat\beta_1 \sum x_i^2 = \sum x_i y_i$

$\hat\beta_0 = \bar y - \hat\beta_1 \bar x, \quad \hat\beta_1 = \dfrac{s_{xy}}{s_{xx}}$

$e_i = y_i - \hat y_i, \quad \sum e_i = \sum x_i e_i = 0$

$\mathrm{SST} = \sum (y_i - \bar y)^2 = \sum (y_i - \hat y_i)^2 + \sum (\hat y_i - \bar y)^2$
$= \mathrm{SSE} + \mathrm{SSR}$

$\mathrm{SST} = s_{yy}, \quad \mathrm{SSR} = \hat\beta_1^2 s_{xx}$

$R^2 = \dfrac{\mathrm{SSR}}{\mathrm{SST}} = \dfrac{\hat\beta_1^2 s_{xx}}{s_{yy}} = \dfrac{s_{xy}^2}{s_{xx}s_{yy}} = \gamma^2$

$T_0 = \dfrac{\hat\theta - \theta_0}{\sqrt{\widehat{\mathrm{Var}}(\hat\theta)}}, \quad \hat\theta \pm \mathrm{t}_{\alpha/2}(\mathrm{df})\sqrt{\widehat{\mathrm{Var}}(\hat\theta)}$

$\widehat{\sigma^2} = \dfrac{\mathrm{SSE}}{n-2} = \mathrm{MSE}, \quad \mathrm{SSE}/\sigma^2 \sim \chi^2(n-2)$

$\mathbb{E}(\chi^2(r)) = r, \quad \mathrm{Var}(\chi^2(r)) = 2r$

$\hat\beta_1 \sim \mathcal{N}\left(\beta_1, \dfrac{\sigma^2}{s_{xx}}\right)$

$\hat\beta_0 \sim \mathcal{N}\left(\beta_0, \sigma^2\left(\dfrac{1}{n} + \dfrac{\bar x^2}{s_{xx}}\right)\right)$

$\hat\mu_{y\cdot x} \sim \mathcal{N}\left(\beta_0 + \beta_1 x, \sigma^2\left(\dfrac{1}{n} + \dfrac{(x-\bar x)^2}{s_{xx}}\right)\right)$

$e_i \sim \mathcal{N}\left(0, \sigma^2\left(1 - \dfrac{1}{n} - \dfrac{(x_i - \bar x)^2}{s_{xx}}\right)\right)$

Box-Cox 변환

$$z_i^{(\lambda)} = \begin{cases} \dfrac{y_i^\lambda - 1}{\lambda}\dfrac{1}{\left(\prod y_i^{\lambda-1}\right)^{1/n}}, & \lambda \neq 0 \\ \log y_i \cdot \left(\prod y_i\right)^{1/n}, & \lambda = 0 \end{cases}$$

측정오차

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad x_i^* = x_i + \delta_i$

$\delta_i \sim \mathcal{N}(0, \sigma_\delta^2) \perp \epsilon_i, \quad \mathrm{data}: (y_i, x_i^*)$

$\mathrm{Assumption}: \dfrac{1}{n}\sum (x_i - \bar x)^2 \xrightarrow{p} \eta_x^2$

$\hat\beta_1 = \dfrac{s_{x^*y}}{s_{x^*x^*}} \xrightarrow{p} \dfrac{\eta_x^2}{\eta_x^2 + \sigma_\delta^2}\beta_1$

회귀선의 비교

$y_{1j} = \beta_{10} + \beta_{11}x_{1j} + \epsilon_{1j}, \quad \epsilon_{1j} \sim \mathcal{N}(0, \sigma^2)$

$y_{2j} = \beta_{20} + \beta_{21}x_{2j} + \epsilon_{2j}, \quad \epsilon_{2j} \sim \mathcal{N}(0, \sigma^2)$

$(\mathrm{H}_0: \beta_{11} = \beta_{21})$

$\widehat{\sigma^2} = \dfrac{\mathrm{SSE}_1 + \mathrm{SSE}_2}{(n_1 - 2) + (n_2 - 2)}$

T statistics of $\hat\beta_{11} - \hat\beta_{21} \overset{\mathrm{H}_0}{\sim} \mathcal{N}(0, \mathrm{Var}(\hat\beta_{11}) + \mathrm{Var}(\hat\beta_{21}))$

$(\mathrm{H}_0: \beta_{10} = \beta_{20}, \beta_{11} = \beta_{21})$

$\mathrm{SSE}_F = \mathrm{SSE}_1 + \mathrm{SSE}_2, \quad \mathrm{df}_F = n_1 + n_2 - 4$

$\mathrm{SSE}_R = \sum_i \sum_j (y_{ij} - \hat\alpha_0 - \hat\alpha_1 x_{ij})^2, \mathrm{df}_R = n_1 + n_2 - 2$

$\mathrm{F} = \dfrac{(\mathrm{SSE}_R - \mathrm{SSE}_F)/(\mathrm{df}_R - \mathrm{df}_F)}{\mathrm{SSE}_F/\mathrm{df}_F}$

$\overset{H_0}{\sim} \mathcal{F}(2, n_1 + n_2 - 4)$

이차형식

Under $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{V}), A = A^T$

1. $\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} \sim \chi^2(\mathrm{r}(\boldsymbol{A}), \frac{1}{2}\boldsymbol{\mu}^T\boldsymbol{A}\boldsymbol{\mu}) \Leftrightarrow \boldsymbol{A}\boldsymbol{V} = (\boldsymbol{A}\boldsymbol{V})^2$
2. $\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} \perp \boldsymbol{B}\boldsymbol{y} \Leftrightarrow \boldsymbol{B}\boldsymbol{V}\boldsymbol{A} = \boldsymbol{O}$
3. $\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} \perp \boldsymbol{y}^T\boldsymbol{B}\boldsymbol{y} \Leftrightarrow \boldsymbol{B}\boldsymbol{V}\boldsymbol{A} = \boldsymbol{O}$ (Under $\boldsymbol{B}^T = \boldsymbol{B}$)
4. $\mathbb{E}(\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y}) = \mathrm{tr}(\boldsymbol{A}\boldsymbol{V}) + \boldsymbol{\mu}^T\boldsymbol{A}\boldsymbol{\mu}$ (Not normal,Any A)
5. $\mathrm{Cov}(\boldsymbol{y}, \boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y}) = \boldsymbol{V}(\boldsymbol{A} + \boldsymbol{A}^T)\boldsymbol{\mu}$ (Not normal,Any A)

Under $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{V}), \boldsymbol{A}_j, \boldsymbol{A} = \sum \boldsymbol{A}_j$ are symmetric

$$\begin{cases} \boldsymbol{y}^T\boldsymbol{A}_j\boldsymbol{y} \sim \chi^2(r(\boldsymbol{A}_j), \frac{1}{2}\boldsymbol{\mu}^T\boldsymbol{A}\boldsymbol{\mu}) \\ \boldsymbol{y}^T\boldsymbol{A}_j\boldsymbol{y} \text{ are independent} \\ \boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} \sim \chi^2(r(\boldsymbol{A}), \frac{1}{2}\boldsymbol{\mu}^T\boldsymbol{A}\boldsymbol{\mu}) \end{cases}$$

$$\Leftrightarrow \mathrm{I}: \begin{cases} \text{Two of (a), (b), (c) hold} \\ \text{(a) } \boldsymbol{A}_j\boldsymbol{V} \text{ is idempotent} \\ \text{(b) } \forall i < j,\ \boldsymbol{A}_i\boldsymbol{V}\boldsymbol{A}_j = \boldsymbol{O} \\ \text{(c) } \boldsymbol{A}\boldsymbol{V} \text{ is idempotent} \end{cases}$$

II : (c) and $\mathrm{r}(\boldsymbol{A}) = \sum \mathrm{r}(\boldsymbol{A}_j)$

Multivariate Normal Distribution

$\boldsymbol{X} \overset{d}{=} \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\boldsymbol{Z}$

$f_{\boldsymbol{X}}(\boldsymbol{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2}\exp\left(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\right)$

중회귀

$\boldsymbol{y}_{n\times 1} = \boldsymbol{X}_{n\times p'}\boldsymbol{\beta}_{p'\times 1} + \boldsymbol{\epsilon}_{n\times 1}$

$\boldsymbol{X}^T\boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}^T\boldsymbol{y}, \quad \hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$

$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y} = \boldsymbol{H}\boldsymbol{y}$

$\boldsymbol{H}^2 = \boldsymbol{H}, \boldsymbol{H}^T = \boldsymbol{H}, \quad \boldsymbol{H}\boldsymbol{X} = \boldsymbol{X}, \boldsymbol{H}\boldsymbol{1} = \boldsymbol{1}$

$\boldsymbol{e}^T\boldsymbol{x} = \boldsymbol{0}, \quad \sum_i \boldsymbol{e}_i = \sum_i \boldsymbol{x}_{ij}\boldsymbol{e}_i = \sum_i \hat{\boldsymbol{y}}_i\boldsymbol{e}_i = 0$

$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu} = \boldsymbol{X}\boldsymbol{\beta}, \sigma^2\boldsymbol{I}_n)$

$\mathrm{SST} = \boldsymbol{y}^T\left(\boldsymbol{I}_n - \frac{1}{n}\boldsymbol{J}_n\right)\boldsymbol{y}$

$\dfrac{\mathrm{SST}}{\sigma^2} \sim \chi^2\left(n-1, \frac{1}{2}\boldsymbol{\mu}^T\left(\boldsymbol{I}_n - \frac{1}{n}\boldsymbol{J}_n\right)\boldsymbol{\mu}\right)$

$\mathrm{SSE} = \boldsymbol{y}^T(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}, \ \dfrac{\mathrm{SSE}}{\sigma^2} \sim \chi^2(n-p-1)$

$\mathrm{SSR} = \boldsymbol{y}^T\left(\boldsymbol{H} - \frac{1}{n}\boldsymbol{J}_n\right)\boldsymbol{y}$

$\dfrac{\mathrm{SSR}}{\sigma^2} \sim \chi^2\left(p, \frac{1}{2}\boldsymbol{\mu}^T\left(\boldsymbol{H} - \frac{1}{n}\boldsymbol{J}_n\right)\boldsymbol{\mu}\right)$

$\mathrm{SSE} \perp \mathrm{SSR}, \quad \dfrac{\mathrm{MSR}}{\mathrm{MSE}} \sim \mathcal{F}(p, n-p-1, \lambda)$

$\hat{\boldsymbol{\beta}} \sim \mathcal{N}_n(\boldsymbol{\beta}, (\boldsymbol{X}^T\boldsymbol{X})^{-1}\sigma^2)$

$\widehat{\sigma^2} = \mathrm{MSE} = \dfrac{\mathrm{SSE}}{n-p-1}$

변수의 직교화

$\begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1} \\ \boldsymbol{O} & \boldsymbol{I} \end{bmatrix}\begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix} = \begin{bmatrix} \cdot & \boldsymbol{O} \\ \cdot & \cdot \end{bmatrix}$

$\underset{n\times 1}{\boldsymbol{y}} = \underset{n\times(q+1)(q+1)\times 1}{\boldsymbol{X}_1\boldsymbol{\beta}_1} + \underset{n\times(p-q)(p-q)\times 1}{\boldsymbol{X}_2\boldsymbol{\beta}_2}$

$\begin{bmatrix} \boldsymbol{X}_2^T\boldsymbol{X}_2 & \boldsymbol{X}_2^T\boldsymbol{X}_1 \\ \boldsymbol{X}_1^T\boldsymbol{X}_2 & \boldsymbol{X}_1^T\boldsymbol{X}_1 \end{bmatrix}\begin{bmatrix} \hat{\boldsymbol{\beta}}_2 \\ \hat{\boldsymbol{\beta}}_1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{X}_2^T\boldsymbol{y} \\ \boldsymbol{X}_1^T\boldsymbol{y} \end{bmatrix}$

$\boldsymbol{X}_2^T(\boldsymbol{I}_n - \boldsymbol{H}_1)\boldsymbol{X}_2\hat{\boldsymbol{\beta}}_2 = \boldsymbol{X}_2^T(\boldsymbol{I}_n - \boldsymbol{H}_1)\boldsymbol{y}$

$\boldsymbol{X}_2^T(\boldsymbol{I}_n - \boldsymbol{H}_1)^2\boldsymbol{X}_2\hat{\boldsymbol{\beta}}_2 = \boldsymbol{X}_2^T(\boldsymbol{I}_n - \boldsymbol{H}_1)^2\boldsymbol{y}$

Let $\boldsymbol{X}_{2\cdot 1} = (\boldsymbol{I}_n - \boldsymbol{H}_1)\boldsymbol{X}_2$

$\boldsymbol{X}_{2\cdot 1}$은 $\boldsymbol{X}_2 = \boldsymbol{X}_1\boldsymbol{\gamma}_1 + \boldsymbol{\epsilon}$ 의 잔차와 동일

$\boldsymbol{X}_{2\cdot 1}^T\boldsymbol{X}_{2\cdot 1}\hat{\boldsymbol{\beta}}_2 = \boldsymbol{X}_{2\cdot 1}^T\boldsymbol{y}$

$\hat{\boldsymbol{\beta}}_2$은 $\boldsymbol{y} = \boldsymbol{X}_{2\cdot 1}\boldsymbol{\gamma}_2 + \boldsymbol{\epsilon}$ 의 $\hat{\boldsymbol{\gamma}}_2$과 동일

$\hat{\boldsymbol{\beta}}_2$은 $\boldsymbol{e} = \boldsymbol{X}_{2\cdot 1}\boldsymbol{\gamma}_2 + \boldsymbol{\epsilon}$ 의 $\hat{\boldsymbol{\gamma}}_2$과 동일

Gauss-Markov 정리

$\mathbb{E}(\boldsymbol{\epsilon}) = 0, \mathrm{Var}(\boldsymbol{\epsilon}) = \sigma^2\boldsymbol{I}, X$ 가 full rank일 때,

$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$ is Best Linear Unbiased Est.

Let $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} + \boldsymbol{A}\boldsymbol{y}$, $\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2\boldsymbol{A}\boldsymbol{A}^T$

$\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}})$ positive definite

$\boldsymbol{C}\boldsymbol{\beta} = \boldsymbol{m}$ 의 검정

$\tilde{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}}\left\{(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^T(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) \mid \boldsymbol{C}\boldsymbol{\beta} = \boldsymbol{m}\right\}$

$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}}\left\{(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^T(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})\right\}$

$\tilde{\boldsymbol{\beta}} = \{I_{p+1} - (X^TX)^{-1}C^T[C(X^TX)^{-1}C^T]^{-1}C\}\hat{\boldsymbol{\beta}}$
$+ (X^TX)^{-1}C^T[C(X^TX)^{-1}C^T]^{-1}\boldsymbol{m}$

$\mathrm{SSE}_R = \mathrm{SSE}(\tilde{\boldsymbol{\beta}}) = \mathrm{SSE}(\hat{\boldsymbol{\beta}}) + Q = \mathrm{SSE}_F + Q$

$Q = (\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{m})^T(\boldsymbol{C}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{C}^T)^{-1}(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{m})$

$\dfrac{Q}{\sigma^2} \sim \chi^2(r(\boldsymbol{C}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{C}^T) = k, \lambda)$

$\mathrm{SSE}(\hat{\boldsymbol{\beta}}) \sim \chi^2(n-p-1), \quad \mathrm{SSE}(\hat{\boldsymbol{\beta}}) \perp Q$

$F_0 = \dfrac{Q/k}{\mathrm{SSE}(\hat{\boldsymbol{\beta}})/(n-p-1)} \sim \mathcal{F}(k, n-p-1, \lambda)$

추가제곱합

$SS(\beta_0) = \boldsymbol{y}^T\frac{1}{n}\boldsymbol{J}_n\boldsymbol{y}$

$\boldsymbol{\beta}_1 = (\beta_0, \dots \beta_q)^T, \boldsymbol{\beta}_2 = (\beta_{q+1}, \dots \beta_p)^T$

$SS(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1) = \mathrm{SSR}_2 - \mathrm{SSR}_1 = \boldsymbol{y}^T(\boldsymbol{H} - \boldsymbol{H}_1)\boldsymbol{y}$

$SS(\boldsymbol{\beta}_1) = \boldsymbol{y}^T\boldsymbol{H}_1\boldsymbol{y}$

$(\boldsymbol{H} - \boldsymbol{H}_1)^2 = (\boldsymbol{H} - \boldsymbol{H}_1), \ r(\boldsymbol{H} - \boldsymbol{H}_1) = p - q$

$\dfrac{SS(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1)}{\sigma^2} \sim \chi^2(p-q, \lambda)$

$\lambda = \boldsymbol{\beta}_2^T\left[\boldsymbol{X}_2^T\boldsymbol{X}_2 - \boldsymbol{X}_2^T\boldsymbol{H}_1\boldsymbol{X}_2\right]\boldsymbol{\beta}_2/(2\sigma^2)$

$\mathrm{SSE} \perp SS(\boldsymbol{\beta}_2|\boldsymbol{\beta}_1)$

Joint Confidence Region

$\boldsymbol{y} = \beta_0 + \beta_1\boldsymbol{x}_1 + \cdots + \beta_p\boldsymbol{x}_p + \boldsymbol{\epsilon}, \quad P(\beta \in R) \geq 1 - \alpha$

$(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T(\boldsymbol{X}^T\boldsymbol{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\sigma^2 \sim \chi^2(p+1), \quad \mathrm{SSE} \perp \hat{\boldsymbol{\beta}}$

$R = \left\{\boldsymbol{\beta} : \dfrac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T(\boldsymbol{X}^T\boldsymbol{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/(p+1)}{\mathrm{SSE}/(n-p-1)} \leq F_\alpha(p+1, n-p-1)\right\}$

Simultaneous Confidence Interval

$\hat{\boldsymbol{\beta}}_j \sim \mathcal{N}(\boldsymbol{\beta}_j, ((\boldsymbol{X}^T\boldsymbol{X})^{-1})_{(j+1)\,(j+1)}\sigma^2)$

β_j의 신뢰도가 서로 동일, 각각의 cartessian product

1. Bonferroni Method

$A_j = [\beta_j | \hat\beta_j \pm t_{\alpha/2(p+1)}(n-p-1)\sqrt{\widehat{\mathrm{Var}}(\hat\beta_j)}]$

$P\left(\bigcap_{j=0}^p A_j\right) = 1 - P\left(\bigcup_{j=0}^p A_j^c\right) \geq 1 - \sum_{j=0}^p P\left(A_j^c\right)$

$= 1 - \sum_{j=0}^p \dfrac{\alpha}{p+1} = 1 - \alpha$

2. Scheffe Method

$c^2 = (p+1)\mathcal{F}_\alpha(p+1, n-p-1)$

$\forall \boldsymbol{a}, P\left(\boldsymbol{a}^T\boldsymbol{\beta} \in \boldsymbol{a}^T\hat{\boldsymbol{\beta}} \pm c\sqrt{\boldsymbol{a}^T(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{a}\mathrm{MSE}}\right) \geq 1 - \alpha$

회귀모형의 비교검정

$\underset{(n_1\times 1)}{\boldsymbol{y}_1} = \underset{(n_1\times p')(p'\times 1)}{\boldsymbol{x}_1\boldsymbol{\beta}_1} + \underset{(n_1\times 1)}{\boldsymbol{\epsilon}_1}$

$\underset{(n_2\times 1)}{\boldsymbol{y}_2} = \underset{(n_2\times p')(p'\times 1)}{\boldsymbol{x}_2\boldsymbol{\beta}_2} + \underset{(n_2\times 1)}{\boldsymbol{\epsilon}_2}$

1. F statistics

$\boldsymbol{X} = (\boldsymbol{x}_1^T\ \boldsymbol{x}_2^T)^T, \quad \boldsymbol{y} = (\boldsymbol{y}_1^T\ \boldsymbol{y}_2^T)^T$

$\mathrm{SSE}_F = \boldsymbol{y}^T\begin{bmatrix} \boldsymbol{I} - \boldsymbol{H}_1 & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{I} - \boldsymbol{H}_2 \end{bmatrix}\boldsymbol{y}$

$\mathrm{SSE}_R = \boldsymbol{y}^T\boldsymbol{H}\boldsymbol{y}, \quad \boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$

$\mathrm{df}_F = n_1 + n_2 - 2(p+1), \quad \mathrm{df}_R = n_1 + n_2 - (p+1)$

2. Hotelling's T statistics

$\hat\beta_1 - \hat\beta_2 \sim \mathcal{N}_{p'}(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_1, ((\boldsymbol{X}_1^T\boldsymbol{X}_1)^{-1} + (\boldsymbol{X}_2^T\boldsymbol{X}_2)^{-1})\sigma^2)$

3. Using $\boldsymbol{C}\boldsymbol{\beta} = \boldsymbol{O}$

$\boldsymbol{C}\boldsymbol{\beta} = [\boldsymbol{I}_{p'}\ -\boldsymbol{I}_{p'}]\begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{bmatrix} = \boldsymbol{\beta}_1 - \boldsymbol{\beta}_2$

Misc.

$\dfrac{\partial(\boldsymbol{c}^T\boldsymbol{x})}{\partial\boldsymbol{x}} = \boldsymbol{c}, \quad \dfrac{\partial(\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y})}{\partial\boldsymbol{y}} = (\boldsymbol{A} + \boldsymbol{A}^T)\boldsymbol{y}$

$\boldsymbol{S} = \boldsymbol{A}_{22} - \boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}$

$\boldsymbol{A}^{-1} = \begin{bmatrix} \boldsymbol{A}_{11}^{-1} + \boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{S}^{-1}\boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\boldsymbol{S}^{-1} \\ -\boldsymbol{S}^{-1}\boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1} & \boldsymbol{S}^{-1} \end{bmatrix}$

$|v^tu| \leq \|u\|\|v\|, \quad HX_1 = X_1, \ HH_1 = H_1$